

KANWAR ROBINSON

Software Development Engineer - II

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SUMMARY

AI-Focused Software Engineer with 2+ years of experience delivering **LLM-driven applications** and **cloud-native microservices**. Specialized in **FastAPI**, **LangChain**, and **LangGraph**, with a strong focus on **multi-agent orchestration**, chatbot architectures, and scalable backend systems. Architected an AI assistant that reduced FastAPI response latency by **40 %** and deflected **35 %** of support tickets. Skilled in deploying high-performance services on **Kubernetes** using **Docker**, **MongoDB**, **Redis**, and **Kafka**. Passionate about building intelligent, production-ready AI systems that scale.

CORE COMPETENCIES

- **Backend** : Python3.11, FastAPI, Java17, Spring boot 3
- **AI/ML** : LangChain, LangGraph, Retrieval Augmented Generation (RAG), RedisVectorStore
- **DevOps** : Kubernetes, Docker, Helm, Jenkins, GitHub Actions
- **Data & Messaging** : MongoDB, Redis Stack Server, Apache kafka, Zookeeper

PROFESSIONAL EXPERIENCE

AppViewX, Inc.

Software Development Engineer - II

08/2023 – present
Coimbatore, India

- Worked on developing a customer support chatbot from scratch, serving as both Onboarding Agent and Documentation Assistant, leveraging **LangChain** and **LangGraph** for **multi-agent framework** with **Redis-Stack** for robust conversational memory and **OpenAI LLM's** for advanced natural language processing.
- Developed a **Checkpoint Service** leveraging LangGraph and Redis to store and retrieve **conversational memory** and user context with millisecond-level response times, enhancing the chatbot's personalization and user experience.
- Refactored a latency-critical Python process from a Java-invoked sub-process into a dedicated **FastAPI-based microservice**, deployed within the same **Kubernetes cluster**. This architectural shift reduced API response latency by approximately 40%, significantly **enhancing the chatbot's performance** and scalability.
- Contributed to the development of a **migration framework** for **Redis Vector Store**, enabling seamless transfer of vector embeddings across AppViewX versions to maintain consistency and backward compatibility during feature rollouts.
- Optimized the AppViewX search inventory to manage **20 million records** with **millisecond response** times by implementing advanced query optimization techniques and efficient database design strategies in **MongoDB**.
- Delivered an RBAC (Role-Based Access Control) feature for out-of-the-box solutions, successfully managing multi-project code changes independently under tight deadlines.

Research & Development Intern

01/2023 – 07/2023
Coimbatore, India

- Worked in developing a new feature called **Expiry Alerts** for the CERT+ product to notify users of SSL certificate expirations, utilizing **Java Spring Framework** and visual workflows in **Python** to enhance product functionality.
- Designed and implemented a live data streaming solution for transferring millions of documents between **MongoDB** clusters using **Apache Kafka** with **ZooKeeper**, ensuring high availability and zero data loss.

Talakunchi Networks

07/2020 – 08/2020
Mumbai, India

Cybersecurity & Ethical Hacking Intern

- Worked on identifying common web vulnerabilities based on the **OWASP Top 10**, such as SQL injection and cross-site scripting. Built simple Python scripts to analyze server logs and highlight unusual activity, which helped the team detect issues faster and reduce open vulnerabilities by **60 %**.
- Conducted hands-on penetration tests with **Nmap**, **Burp Suite**, and **Metasploit**, mapping attack surfaces and exploiting misconfigurations to produce a prioritized remediation report that eliminated three critical vulnerabilities during the internship.

EDUCATION

Bachelor of Engineering (Mechanical Engineering)

07/2019 – 05/2023

Coimbatore Institute of Technology, Coimbatore, India [GPA - 8.57/10]

PROJECT HIGHLIGHTS

Agentic AI Assistant

Developed a multi-agent chatbot architecture using **Agentic AI design patterns** with **LangGraph** and **RAG**, enabling dynamic tool use, memory chaining, and goal-driven execution. Built on **FastAPI** and containerized with **Docker**, the system was deployed on **Kubernetes (GKE)** using **Helm** for rollout management, with **OpenTelemetry tracing** and **Prometheus/Grafana** for real-time observability and monitoring.

Motion Planning and Path Tracing of Robotic Arm using MATLAB and RoboAnalyzer

Developed a motion planning and path tracing system for a robotic arm using **MATLAB** and **RoboAnalyze**, enabling precise trajectory generation and optimization. The project focused on simulating and analyzing kinematics to enhance robotic movement efficiency.

Laser Powder Bed Fusion of Aluminium Tin Alloy: Manufacturing, Microstructure, and Mechanical Properties

Investigated the manufacturing, microstructure, and mechanical properties of Aluminium-Tin alloy using Laser Powder Bed Fusion and Casting techniques. Conducted tensile, fatigue, and wear tests to optimize composition for enhanced industrial applications.

AWARDS

Circle of Excellence - AppViewX, Inc.	07/2024
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Awarded the **Circle of Excellence 2023** for tackling key challenges in **CERT+**, including expiry alerts, onboarding streamlining, UI enhancements, infrastructure scaling, and driving product-led growth.

Engineer of the Month - AppViewX, Inc.	12/2024
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Awarded for implementing **Redis Stack Server** to enhance robust conversational memory and maintain the state in the **LLM engine**.

Spot Award - AppViewX, Inc.	11/2023
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Recognized with a Spot Award for efficiently implementing automated Expiry Alert workflows across Jira, Slack, ServiceNow, and CERT+ OOB roles, enhancing workflow automation and operational efficiency.